THE EXPANDING AI LANDSCAPE

A global race is intensifying to develop more powerful and efficient Large Language Models (LLMs), deep-learning systems trained on vast data to process language and images. They form the foundation of Generative AI, an innovation that has revolutionised content creation

GLOBAL MODELS

The 2022 launch of ChatGPT, a GenAI chatbot, unveiled the potential of LLMs to a global audience, sparking a rapidly expanding ecosystem

GPT-4
Launched by OpenAI in 2023, the Generative Pre-Trained Transformer-4 was estimated to be trained on 1.7 trillion parameters*—10 times more values than its previous version, GPT-3

DEEPSEEK R1
An open-source model released in Jan. 2025 by Chinese startup DeepSeek, which claims the cost of training was less than $6 million as compared to over $100 million spent to train GPT-4

LLAMA
A family of models released by Facebook's parent company Meta in 2023, Llama is 'open weight', which is a midway approach between open source and closed systems in terms of transparency and control

MISTRAL
A French competitor, Mistral has a range of LLMs, the latest being Mistral Large 2, released in 2024, with 123 billion parameters

BERT
An early open-source family of models launched by Google starting 2018, based on pioneering research into transformer architecture for artificial neural networks

MADE IN INDIA

India's first set of native language models emerged in late 2023; now there are 17-plus infrastructure GenAI startups developing various kinds of LLMs

BHARATGEN
An Indian government-led initiative to build high-scale LLMs from ground up; 2 billion parameter text models for Hindi and English have been developed and 5 billion parameter models are being developed for 13 Indian languages

SARVAM 1
An open-source 2-billion parameter LLM launched in 2024 by Bengaluru startup Sarvam AI. It has been trained on 10 Indian languages apart from English

KRUTRIM
Family of LLM models launched in 2023 by Ola founder Bhavish Aggarwal, who recently announced investment of Rs 2,000 crore in the AI startup, also named Krutrim

BHARATGPT
LLM launched in 2023 by Bengaluru-based conversational AI start-up CoRover.ai which develops video bot, voice bot and chat bot solutions

EVEREST
Multilingual LLM launched in 2024 by tech startup Hanooman AI

**Parameters refer to the variables that an LLM is trained on; the more the parameters, the more complex patterns it can learn*



"By enabling an ecosystem around its foundational models, BharatGen envisions positioning India as a global leader in Generative AI"

PROF. ABHAY KARANDIKAR, Secretary, Department of Science and Technology

el trained on two billion parameters in Hindi and English while five-billion parameter models are being developed for 13 Indian languages. Parameters refer to the variables on which an AI model is trained; the higher the number of parameters, the bigger the model. In that sense, these are typically small models. For comparison, OpenAI's chatbot ChatGPT was trained on 20 billion parameters while GPT-4 is believed to have had 1.76 trillion parameters.

But that's also where DeepSeek's innovative engineering has turned the world of AI tech on its head—its R1 model not only utilised fewer GPUs but activated only a fraction (37 billion) of its total 671 billion parameters for each token processed. (A token is a basic unit of input processed by an LLM.) This was because DeepSeek made use of a machine learning technique known as a 'mixture of experts'. The analogy that tech specialists point to is how the human brain can carry out complex tasks with very little energy requirement. "This happens because the brain delegates different functionalities and tasks to different parts. For example, vision is processed at the back of the brain, hearing is handled by the temporal lobe...and, amazingly, different parts of the brain can take up another part's job when that part has failed," explains Pushpak Bhattacharyya, a computer scientist at IIT Bombay. "So this idea of delegating tasks and making the models interact with each other goes back to nature itself."

Bhattacharyya too has been pursuing the technique, which he calls the Trinity Model, for the past five years. This method links smaller models for collective intelligence, creating the